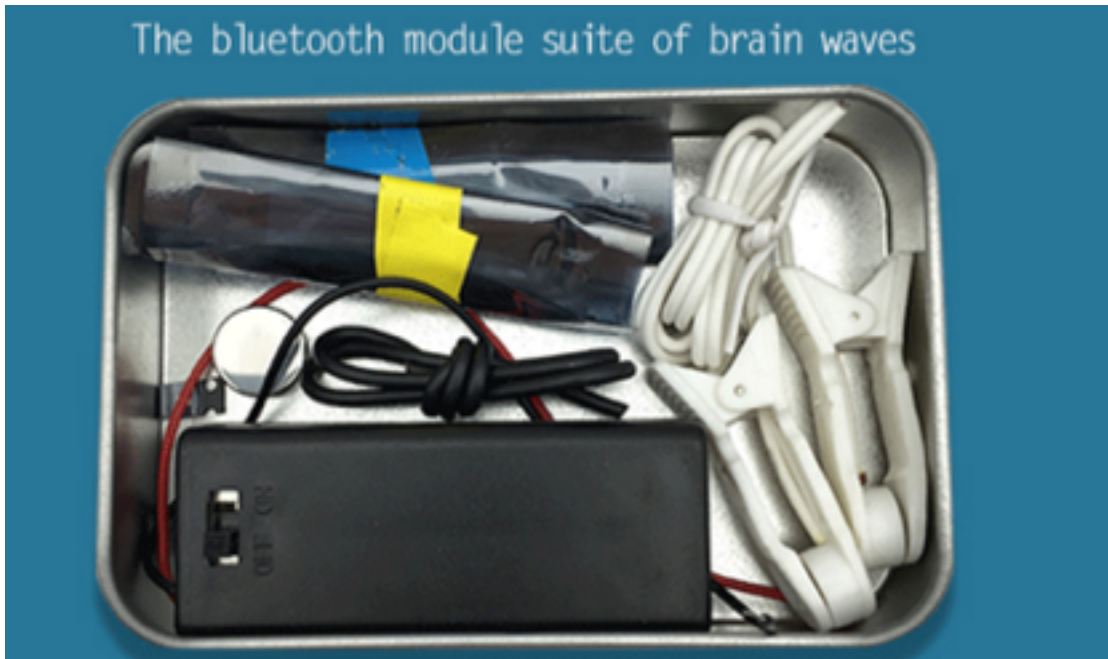


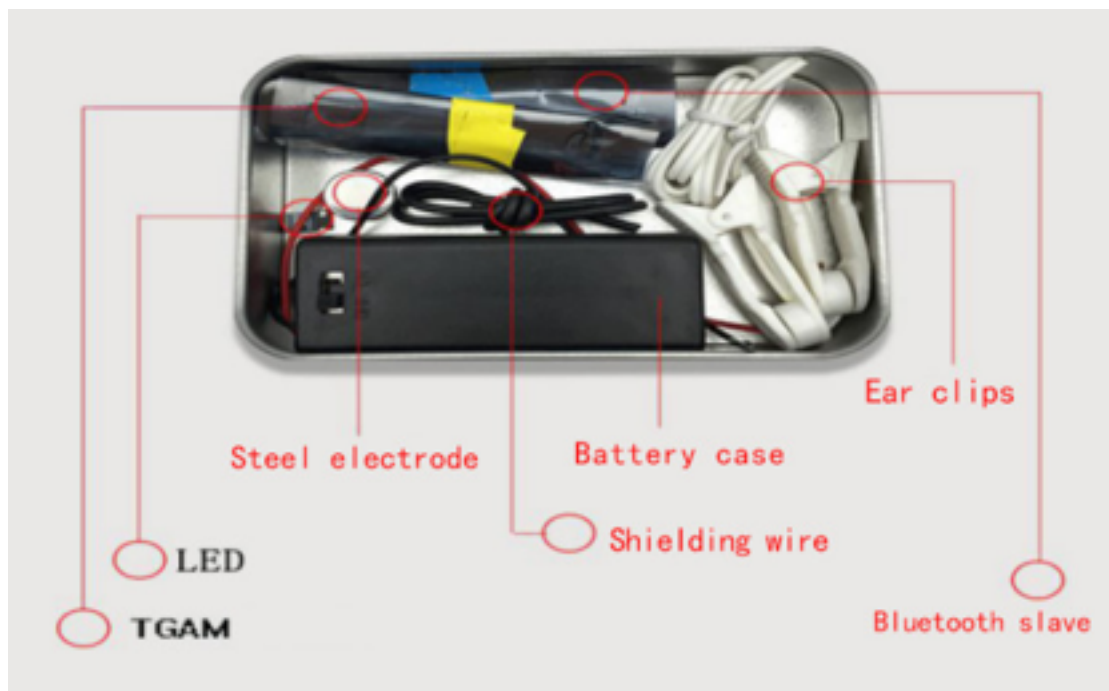
The bluetooth module suite of brain waves



TGAM Starter Kit Soldering & Testing

Material List:

Bluetooth Salve X 1
TGAM X 1
Steel Electrode X 1
EarClips X 1
Shield Wire X 1
LED 6030 X 1



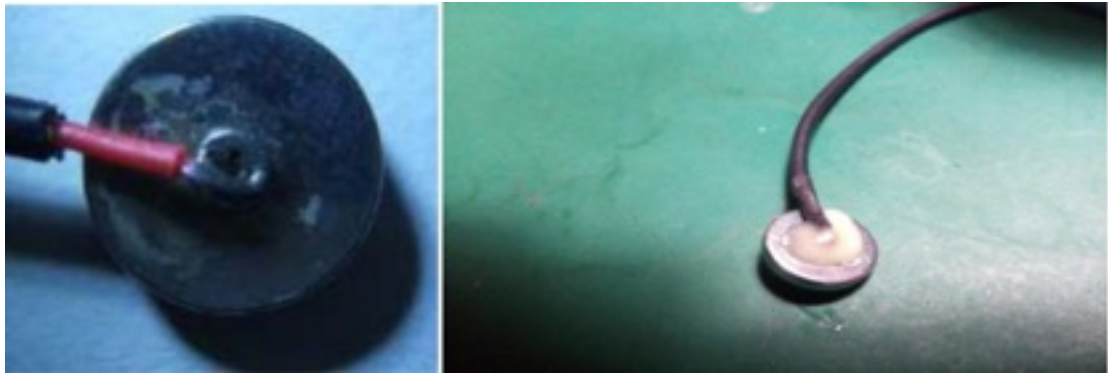
Step1

Strip the black wire and separate the copper shielding and the red core. Remove the copper shielding and the red shell, And stick some tin on the core wire.



Step2

Put heat-shrinkable tube though the core wire, attach the wire to the steel electrode with the backside screw. Heat the tube, and put on the hot-melt adhesive to fix them.



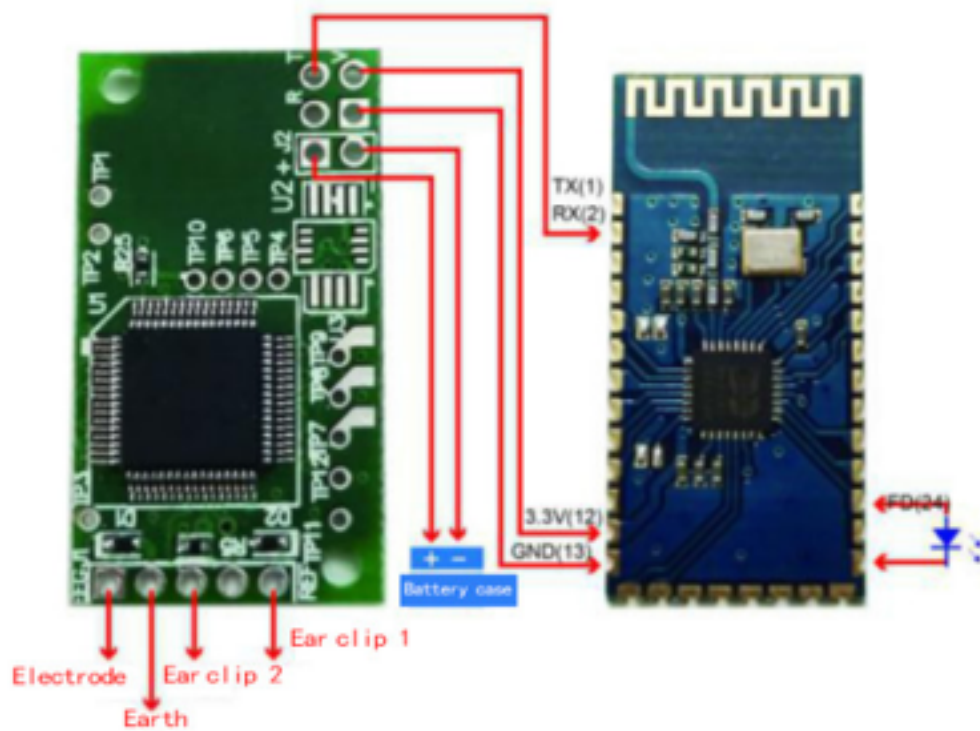
Step3

Solder the LED to the bluetooth slave module. LED green' s part to the “+” pin (pin24) and the other side to the “-” pin (pin26). Please refer to the below picture.

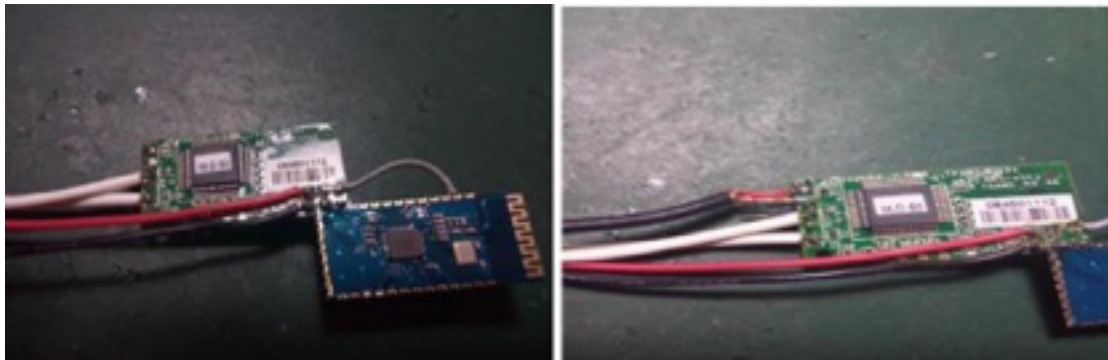


Step4

Add the electrode and earclips to the TGAM. You could find 5 pins on one side of TGAM. Pin1 is for Electrode, Pin2 for the Earth (The copper shielding of the black wire) . Pin3 is for earclip, Pin4 is no need hear. Pin5 is the other earclip. You could find the other 6 pins at the corner, J2 is for the power input (3 volts) , the square pin is for “+” ,the other is “-” . The middle line Rx connects to nothing. The pin next to the Rx is for GND,connect to the pin13 of bluetooth module. Pin Tx connects to the Rx pin (Pin2) on the bluetooth module. Pin V is the power out,connects to the Pin12 of the bluetooth. You could find the detail from the below pictures.

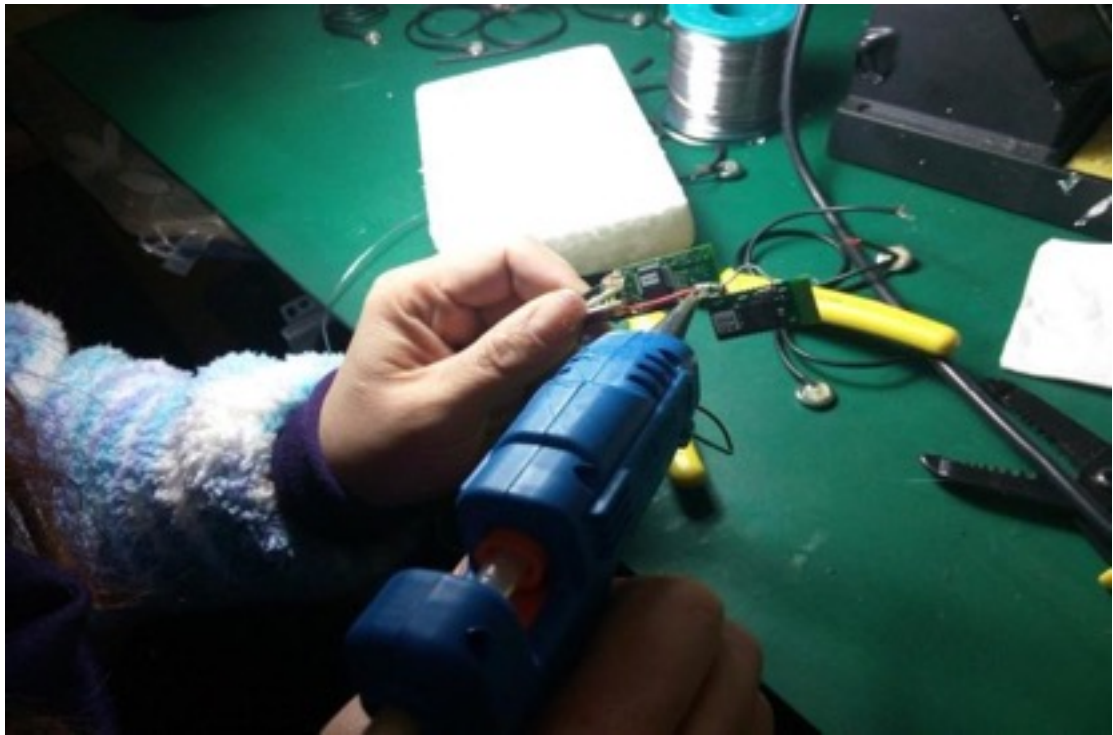


soldering details.



Step5

Add the hot melt glue to all soldering part.



when all works done, the kit has been as below picture.



Step6

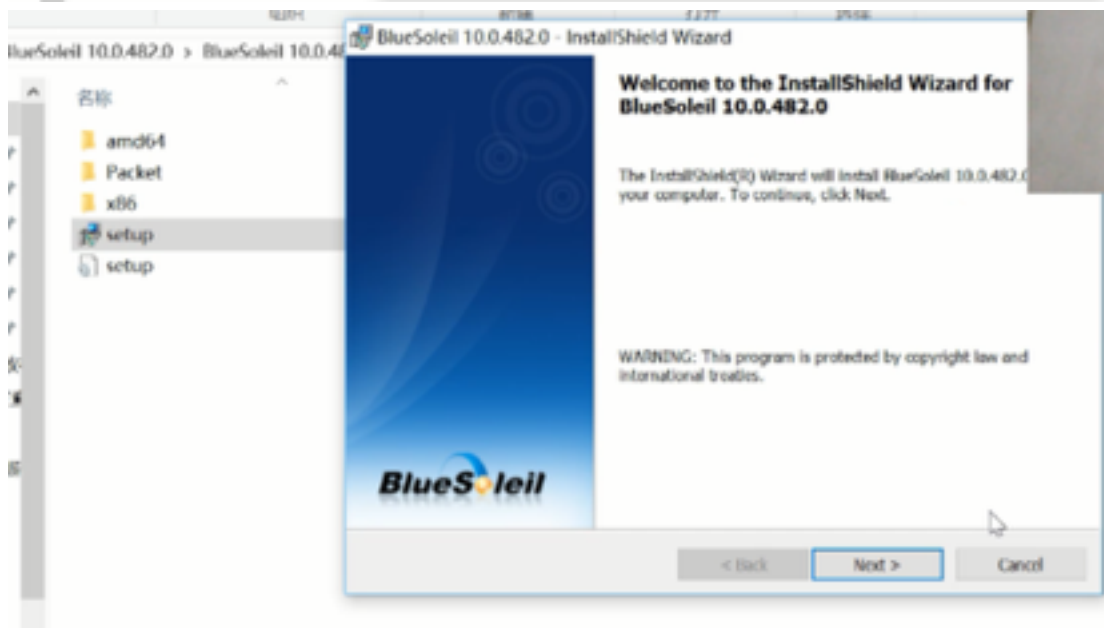
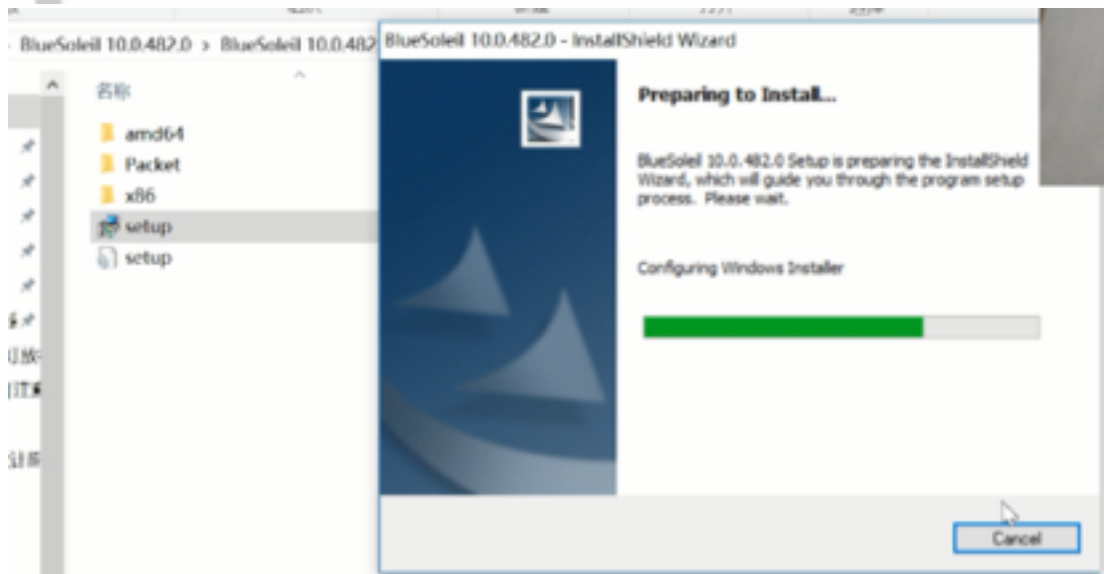
If you have bought our bluetooth dongle (for PC testing), we need to install the bluetooth driver first. Most the bluetooth of the laptop is BLE bluetooth which is not support TGAM bluetooth 2.0 kits.

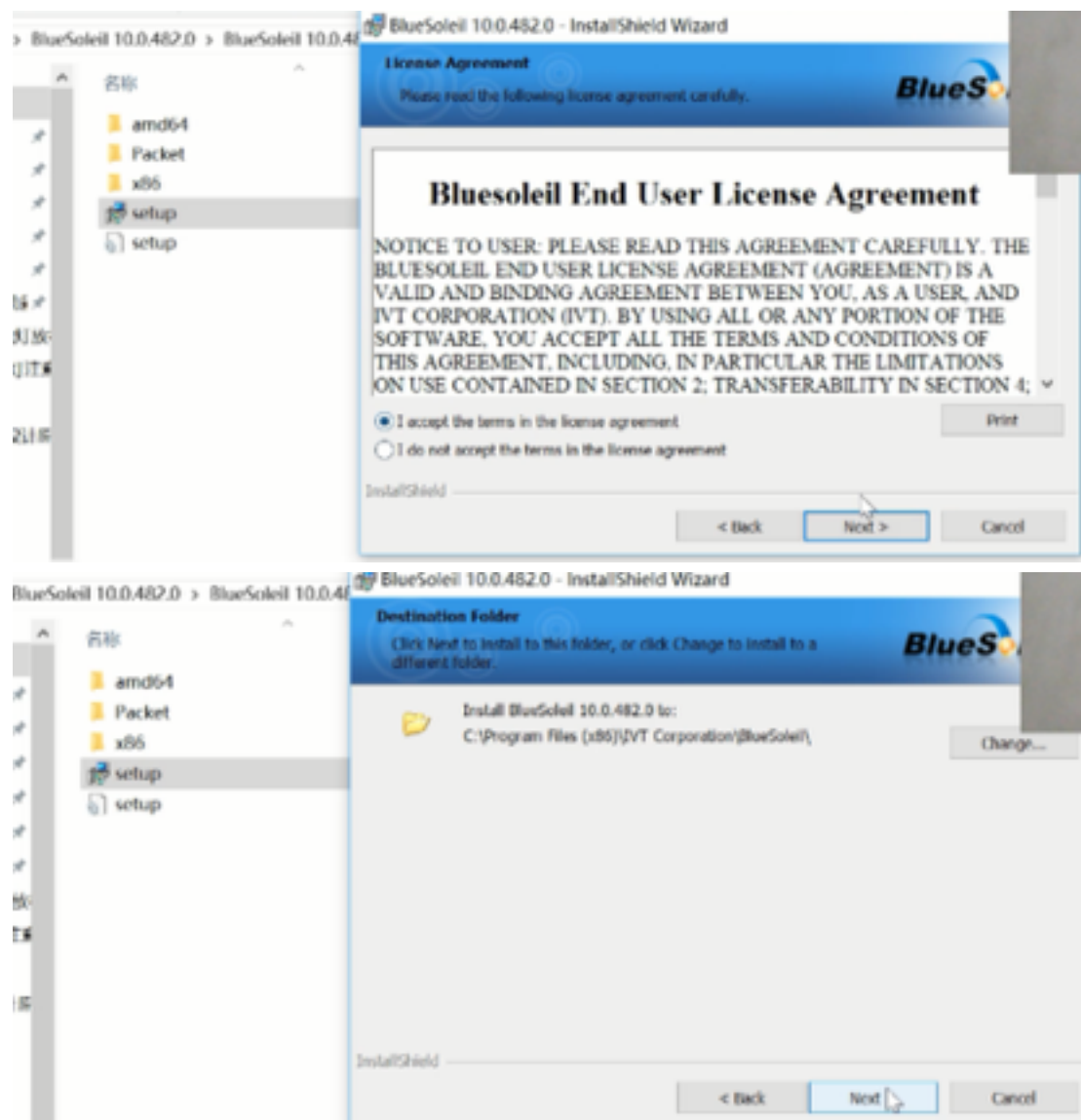
Insert the Bluetooth USB dongle into the laptop' s USB interface.

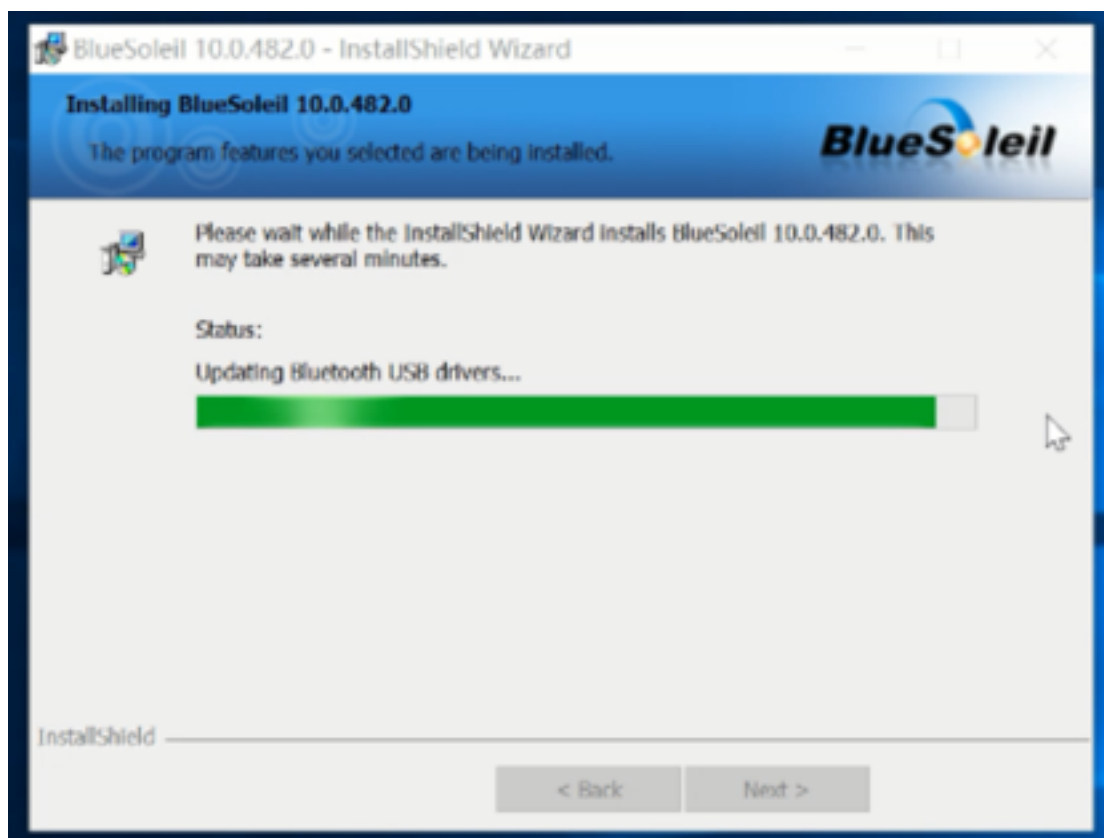
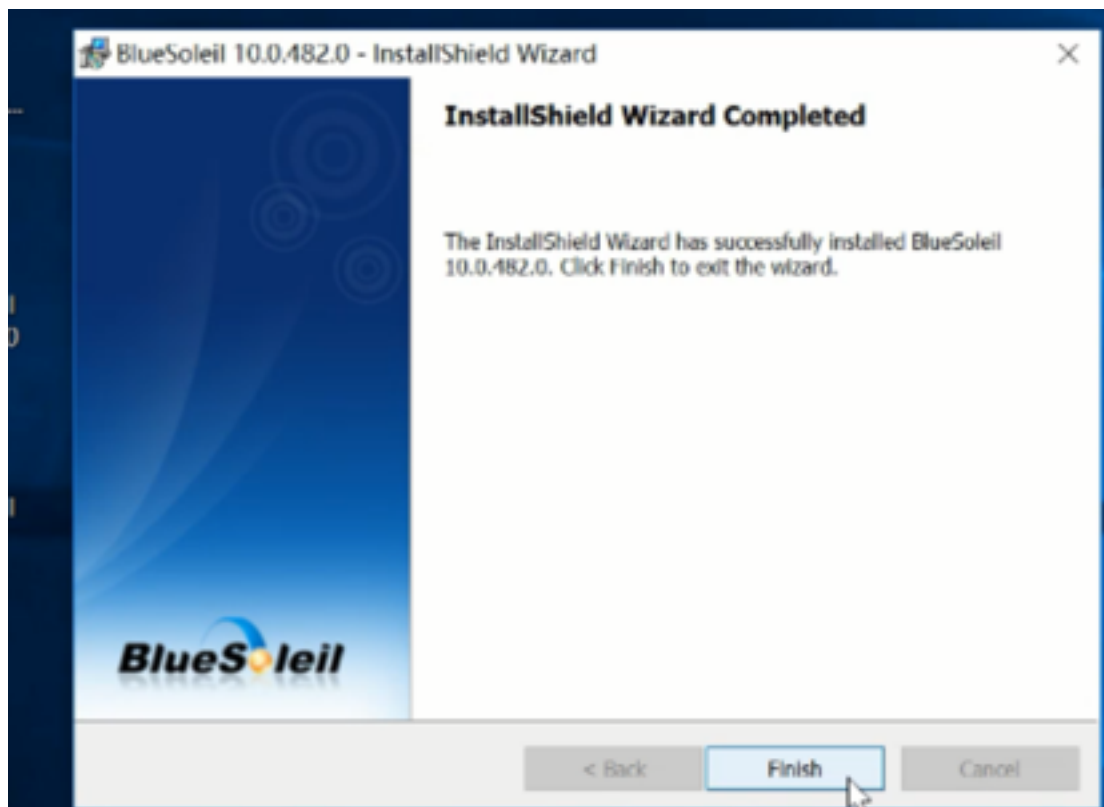


Find and open the documents “BlueSoleil 10.0.4820” , click the “set-up.exe” to install the driver.

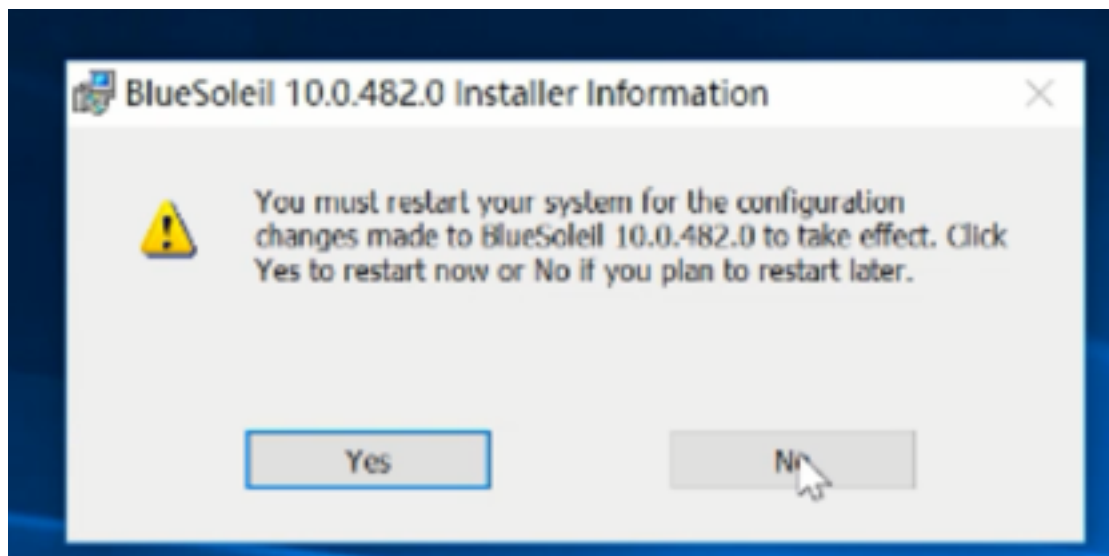
BlueSoleil 10.0.482.0 > BlueSoleil 10.0.482.0 > install







When finished, please restart the laptop.



Then you could find a new bluetooth icon on the desktop. Double-click the icon and open the bluetooth manager, you could find the yellow ball in the middle. Right click and begin to search the bluetooth device. If you could find some device named "Sichiray", right click on the "sichiray" and click "pairing" to pair. When pair successfully, you could find there is a green "chain" icon on your "sichiray" device.



Right click the "sichiray" device again, you could find the serial interface number (such as COM4), please write it down (you will use it next step) and click "connect". If you build up a bluetooth connection with the TGAM kit, you will find the icon has turned into green.

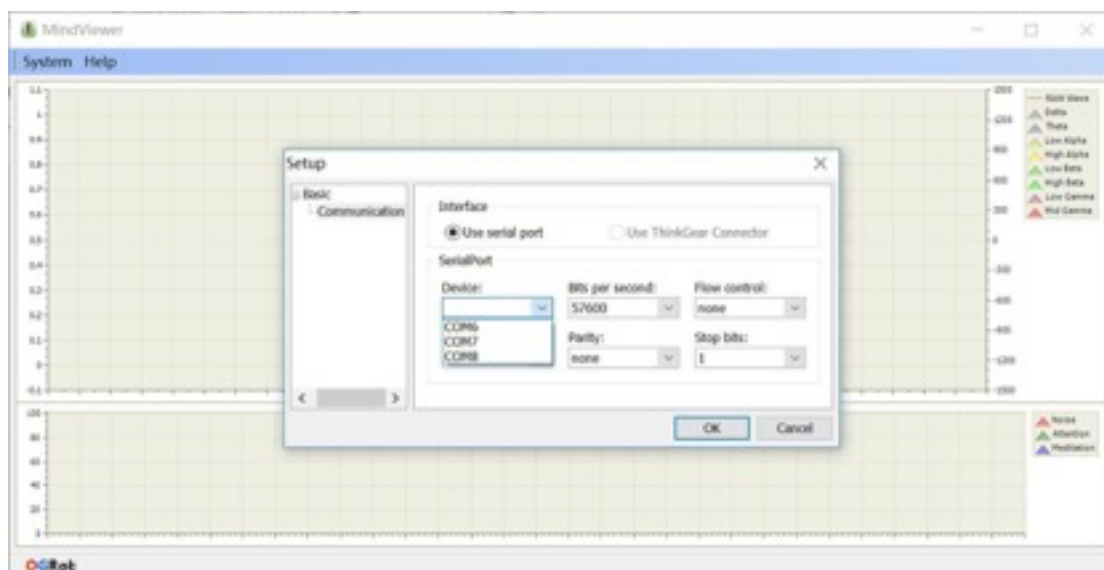


Put on the kit as the below picture, the electrode should be attached on your left forehead and remove all hair between the electrode and skin.

Clip the earclips onto your both earlobes



Go back to the document “TGAM starter Kit” ,you could find a mind-view.exe. Open it. and set the device to the COM interface that you have written down. The baud rate is 57600. Click “Ok” .

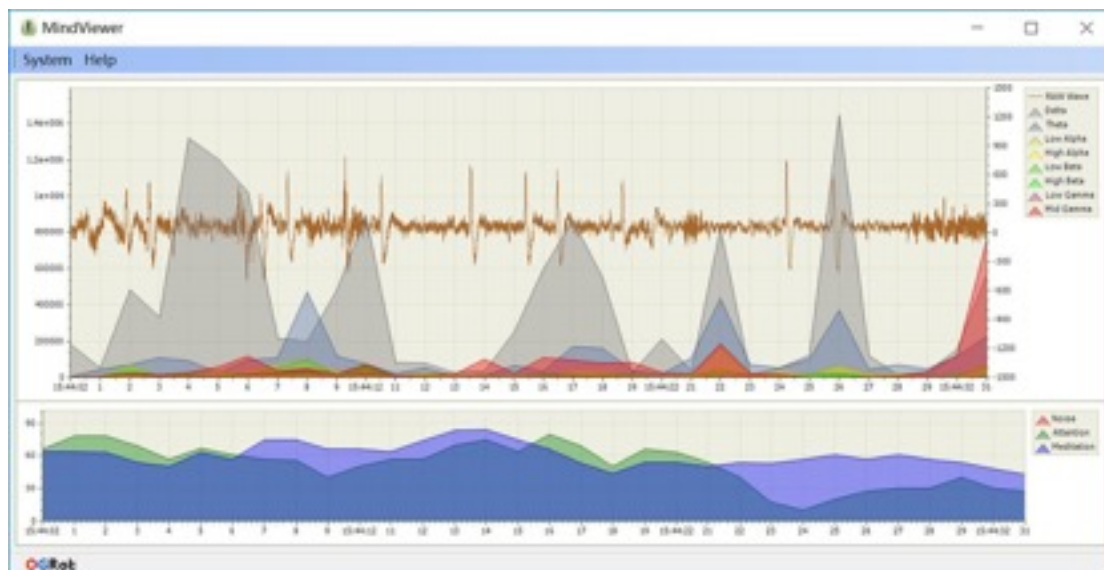


If you find a lot of noise on chart, you need to put on some

clear water on the electrode and ear clips.



The below picture is the out result. That means the kit works well.



Step7

If you have not purchased our bluetooth dongle, you could test your kit on an-droid mobile phone. Please find the eggID.apk. And install the APK file on you phone.



Turn on the bluetooth function in the Setting. Search and pair the device named “sichiray”. The pairing code is 0000.



After pairing successfully, put on the kit on head (Refer to Step6, how to put on the kit) and open the App. If everything is Okay. You could finally see the data flow as the below.

